Mark Moeller

mmoeller@holycross.edu

Education

Cornell University, Ithaca NY 2020-2026
PhD in Computer Science, in the Programming Languages research area, coadvised by Nate Foster and Alexandra Silva

University of Pennsylvania, Philadelphia PA 2012-2013
MSE, Computer and Information Science

United States Naval Academy, Annapolis MD 2008-2012
BS, Mathematics (Honors) and Computer Science

Teaching Experience

College of the Holy Cross (Asst. Prof.)

CSCI226 Computer Systems and Organization	Fall ‘26
CSCI235 Analysis of Algorithms	Fall ‘26

Cornell University (Instructor)

CS2800 Mathematical Foundations of Computing	Fall ‘25

Cornell University (Teaching Assistant)

CS3410 Computer Architecture	Fall ‘20, Spring ‘21

United States Naval Academy (Instructor)

SI413 Programming Languages	Fall ‘18
SI340 Theory of Computing	Fall ‘18, ‘19[1]
IC221 Systems Programming	Spring ‘19
SI335 Computer Algorithms	Spring ‘19[1], ‘20[1]

Note 1. Course coordinator

Professional Experience

Assistant Professor 2026-Present
The College of the Holy Cross, Worcester MA

Research Engineering Intern Summers 2022, 2025
Galois Inc., Portland OR

- Pursued research in network verification and compilers on client-billable projects.

Instructor 2018-2020
United States Naval Academy, Annapolis MD

- Taught courses for Computer Science majors as instructor of record.
- Served as academic advisor for 18 students.

Publications

- Thomas Lu, Qiancheng Fu, Kevin Batz, Oliver Bøving, Tiago Ferreira, Mark Moeller, Nate Foster, Alexandra Silva. A Fast Quantitative Analyzer for NetKAT. In POPL 2027.
- Mark Moeller, Tiago Ferreira, Thomas Lu, Nate Foster, Alexandra Silva. Active Learning of Symbolic NetKAT Automata. In PLDI 2025.
- Mark Moeller + Jules Jacobs, Olivier Savary Belanger, David Darais, Cole Schlesinger, Steffen Smolka, Nate Foster, Alexandra Silva. KATch: A Fast Symbolic Verifier for NetKAT. In PLDI 2024.
- Mark Moeller, Thomas Wiener, Alaia Solko-Breslin, Caleb Koch, Nate Foster, Alexandra Silva. Automata Learning with an Incomplete Teacher. In ECOOP 2023.

Invited Talks

- "Automata Learning for Network Verification." Worcester Polytechnic Institute (WPI) Computer Science Colloquium. November 13, 2026

Service

College Service

- Holy Cross Admissions Open House (Computer Science): November 8, 2026.

Seminar Organization

- Upstate Programming Languages Seminar, August 2025. https://www.cs.cornell.edu/upstate-pl/.
- Programming Languages Discussion Group (PLDG), Cornell's PL Research Seminar, 2022-2024. E.g., https://pl.cs.cornell.edu/pldg/2023sp/

Fellowship

John J. McMullen Fellow 2012
Awarded competitively for complete funding of Master's work, 2012-2013.